Self Check 1.2 • Topics 1.4 and 1.5

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work, start from a physics principle or equation, and justify your answers. A calculator and the AP Physics 1 equation sheet are allowed. Use $g = 10\,\text{m/s}^2$ and ignore air resistance unless stated.

Question 1

A boat moves straight across a river at 6 m/s relative to the water. The current is 8 m/s. What is the boat's speed relative to the bank?

(A) 14 m/s
(B) 2 m/s
(C) 10 m/s
(D) 48 m/s

Solution
Answer: (C). The two velocities are perpendicular: $\sqrt{6^2 + 8^2} = 10\,\text{m/s}$.

Question 2

Rain falls straight down at 8 m/s. You run at 6 m/s. At what speed does the rain hit you, in your reference frame?

(A) 2 m/s
(B) 10 m/s
(C) 14 m/s
(D) 8 m/s

Solution
Answer: (B). Relative to you the rain also moves backward at 6 m/s: $\sqrt{8^2 + 6^2} = 10\,\text{m/s}$.

Question 3

A ball is thrown horizontally from the top of a building 45 m tall. How long does it take to reach the ground?

(A) 3 s
(B) 4.5 s
(C) 9 s
(D) 2 s

Solution
Answer: (A). $45 = \frac{1}{2}(10)t^2 \Rightarrow t^2 = 9$, $t = 3\,\text{s}$.

Question 4

During the flight of a projectile (no air resistance), the horizontal component of its velocity

(A) increases
(B) decreases
(C) stays constant
(D) is zero at the top

Solution
Answer: (C). There is no horizontal acceleration.

Question 5

A projectile is launched at 40 m/s at 30° above the horizontal. What is its maximum height above the launch point?

(A) 20 m
(B) 80 m
(C) 40 m
(D) 10 m

Solution
Answer: (A). $v_{0y} = 40\sin 30^\circ = 20\,\text{m/s}$; $h = \frac{20^2}{2(10)} = 20\,\text{m}$.

Question 6

A force of 10 N acts at 37° above the horizontal ($\sin 37^\circ = 0.6$). What is its vertical component?

(A) 8 N
(B) 6 N
(C) 10 N
(D) 4 N

Solution
Answer: (B). $10\sin 37^\circ = 6\,\text{N}$.

Question 7 (Free response)

A plane flies horizontally at 60 m/s at a height of 500 m and releases a package.

(a) How long does the package take to reach the ground?
(b) How far does it travel horizontally?
(c) Where is the plane when the package lands? Explain using reference frames.

Solution
(a) $500 = \frac{1}{2}(10)t^2 \Rightarrow t = 10\,\text{s}$. (b) $60 \times 10 = 600\,\text{m}$. (c) Directly above it: the package keeps the plane's horizontal velocity, so in the plane's frame it falls straight down.